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Assessment	CO4 Assessment Tool 1 — Test Case Design Assignment
Assigned System	AI-Based Smart Cold Chain Logistics Monitoring and Optimization Platform

1. Requirements Identification

Before scenarios or test cases can be written, the requirements of the AI-Based Smart Cold Chain Logistics Platform were reviewed and split into functional requirements (what the system must do) and non-functional requirements (how well it must do it). Both types drive different kinds of test cases below.

Functional Requirements (FR)

ID	Requirement
FR-1	System shall record temperature and humidity readings from IoT sensors at regular intervals.
FR-2	System shall track and update shipment GPS location in real time.
FR-3	System shall predict refrigeration equipment failure using historical telemetry.
FR-4	System shall recommend an optimized delivery route based on distance, traffic and risk.
FR-5	System shall generate an automated alert when a sensor reading breaches a safe threshold.
FR-6	System shall authenticate users before granting access to the dashboard.

Non-Functional Requirements (NFR)

ID	Requirement
NFR-1	Alerts must be delivered within 5 seconds of a threshold breach (performance).
NFR-2	The dashboard must remain usable with at least 500 concurrent shipments tracked (scalability).
NFR-3	Sensor and login data must be transmitted over encrypted channels (security).
NFR-4	The system must remain operational if a single microservice restarts (reliability).
NFR-5	Dashboard screens must be understandable to a non-technical logistics operator (usability).

2. Test Scenario Identification

Each requirement was expanded into one or more concrete test scenarios: realistic situations a tester can walk through end to end. The process used to move from a raw requirement to an executable test case is summarised below.

TEST CASE DESIGN PROCESS

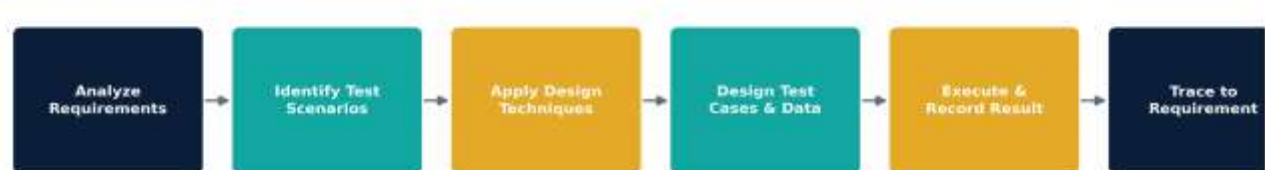


Figure 2.1 — Process followed to derive test cases from requirements

Identified Test Scenarios

Scenario ID	Test Scenario	Related Requirement
TS-01	Verify sensor readings within the safe range are logged without triggering an alert.	FR-1, NFR-1
TS-02	Verify readings at and beyond the safe temperature boundary are handled correctly.	FR-1, FR-5
TS-03	Verify GPS location updates reflect the shipment's real-time position.	FR-2
TS-04	Verify the system behaves correctly when GPS signal is lost mid-transit.	FR-2, NFR-4
TS-05	Verify the failure-prediction model flags a high-risk reefer unit before breakdown.	FR-3
TS-06	Verify prediction behaviour when sensor input data is missing or corrupted.	FR-3
TS-07	Verify the route optimizer proposes the lowest-risk route for a valid shipment.	FR-4
TS-08	Verify alert notifications are generated and delivered within the required time.	FR-5, NFR-1
TS-09	Verify only authorized users can log in and reach the dashboard.	FR-6, NFR-3
TS-10	Verify the system remains available when one microservice is restarted.	NFR-4

3. Test Design Techniques Applied

Four complementary test design techniques were applied so that the test suite catches errors a single technique would miss on its own.

3.1 Equivalence Partitioning — Temperature Input

The temperature input field was split into equivalence classes; one representative value from each class is enough to test that class.

Class	Range / Condition	Representative Value	Expected Handling
Valid	2°C to 8°C (inclusive)	5°C	Accepted, logged as normal
Invalid — too low	< 2°C	-5°C	Rejected / alert triggered
Invalid — too high	> 8°C	15°C	Rejected / alert triggered
Invalid — non-numeric	Text or null sensor payload	"ERR" / null	Input rejected, exception logged

3.2 Boundary Value Analysis — Temperature Thresholds

BVA focuses specifically on the edges of the valid range identified above, since defects cluster at boundaries far more often than in the middle of a range.

BOUNDARY VALUE ANALYSIS — VALID TEMPERATURE RANGE (2°C - 8°C)



Figure 3.1 — Boundary values tested around the 2°C–8°C safe range

3.3 Decision Table — Alert Generation Logic

Alert generation depends on the combination of temperature status and humidity status, so a decision table captures every combination cleanly.

Rule	Temp In Range?	Humidity In Range?	Door Open?	Action
R1	Yes	Yes	No	No alert — log normal
R2	No	Yes	No	Temperature alert
R3	Yes	No	No	Humidity alert
R4	No	No	No	Critical combined alert
R5	Yes	Yes	Yes	Door-open warning
R6	No	—	Yes	Critical alert + door warning

3.4 State Transition Testing — Shipment Lifecycle

Because a shipment moves through a well-defined sequence of states, state transition testing verifies both valid transitions (e.g. In Transit → Delivered) and that invalid transitions (e.g. Delivered → In Transit) are correctly rejected.

STATE TRANSITION DIAGRAM — SHIPMENT LIFECYCLE

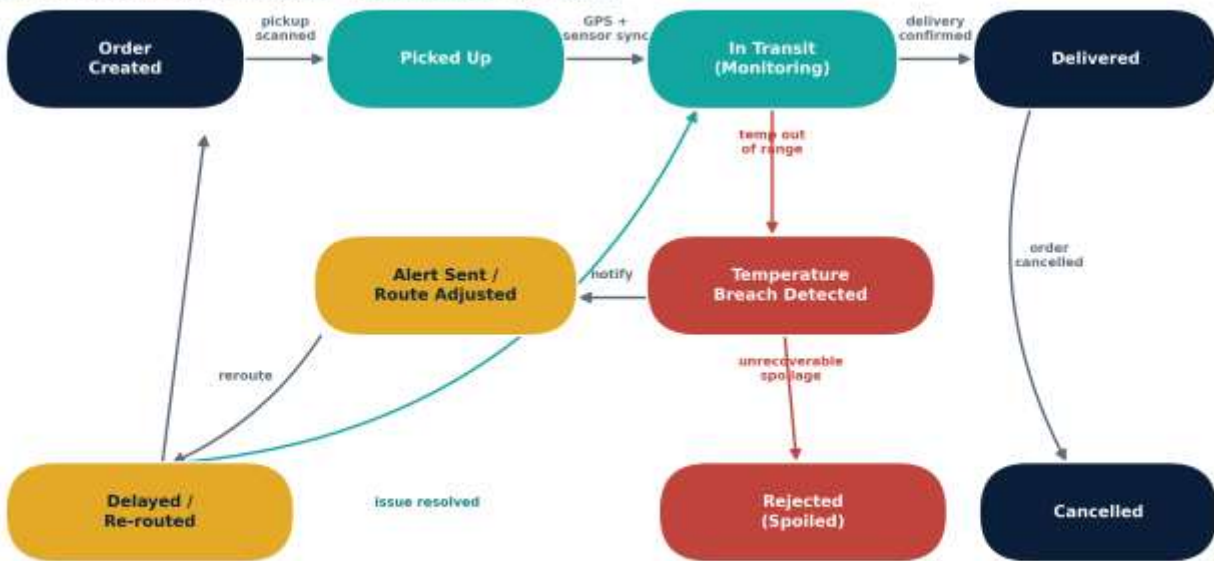


Figure 3.2 — Valid shipment states and transitions used to design state-based test cases

4. Detailed Test Cases

Test cases below cover normal, boundary, invalid, and exceptional conditions across all six functional modules, using the techniques introduced in Section 3. Actual results reflect a first execution pass against a staging build; items marked Fail* indicate defects logged for developer follow-up.

Module A — Temperature & Humidity Monitoring

ID	Scenario	Steps	Test Data	Expected Result	Actual Result	Status
TC-01	Normal reading within safe range	1. Send sensor payload 2. Observe dashboard log	temp=5°C, hum=55%	Reading logged as Normal; no alert	Logged as Normal; no alert	Pass
TC-02	Boundary — minimum valid temperature	1. Send reading at lower limit	temp=2°C	Accepted as valid (pass boundary)	Accepted as valid	Pass
TC-03	Boundary — just below minimum	1. Send reading just under limit	temp=1°C	Rejected / alert triggered	Alert triggered correctly	Pass
TC-04	Boundary — maximum valid temperature	1. Send reading at upper limit	temp=8°C	Accepted as valid (pass boundary)	Accepted as valid	Pass
TC-05	Boundary — just above maximum	1. Send reading just over limit	temp=9°C	Rejected / alert triggered	Alert triggered correctly	Pass
TC-06	Invalid — non-numeric sensor payload	1. Send malformed payload 2. Check error handling	temp="ERR"	Input rejected; exception logged, no crash	System crashed silently, no log entry	Fail*
TC-07	Exceptional — sensor disconnected mid-transit	1. Simulate sensor timeout 2. Wait 60s	no signal for 60s	"Sensor Offline" warning raised	Warning raised as expected	Pass

Module B — GPS Tracking

ID	Scenario	Steps	Test Data	Expected Result	Actual Result	Status
TC-08	Normal GPS location update	1. Send GPS coordinates 2. View map	lat=13.08, lng=80.27	Marker updates to new position	Marker updated correctly	Pass
TC-09	Boundary — coordinate at geofence edge	1. Send location exactly on zone border	boundary lat/lng	Classified as inside delivery zone	Classified as inside zone	Pass
TC-10	Invalid GPS coordinates	1. Send out-of-range coordinates	lat=999, lng=999	Input rejected with validation error	Rejected with validation error	Pass
TC-11	Exceptional — GPS signal lost in transit	1. Simulate signal loss 2. Wait for timeout	no GPS fix for 5 min	"Location Unavailable" shown, last known position retained	Last known position retained correctly	Pass

Module C — AI Failure Prediction

ID	Scenario	Steps	Test Data	Expected Result	Actual Result	Status
TC-12	Normal prediction with sufficient history	1. Feed 30 days telemetry 2. Request prediction	30-day dataset	Risk score returned (0–1) with confidence	Risk score 0.31 returned correctly	Pass
TC-13	Boundary — risk score exactly at alert threshold	1. Feed data producing score = 0.70	score=0.70 (threshold)	Classified as "High Risk" (inclusive)	Classified as High Risk	Pass
TC-14	Invalid — corrupted telemetry input	1. Feed malformed/partial dataset	missing 80% of fields	Graceful fallback / "insufficient data" response	Model returned NaN with no fallback	Fail*
TC-15	Exceptional — prediction service unavailable	1. Stop AI/ML container 2. Request prediction	service down	Dashboard shows fallback message, no crash	Fallback message shown, no crash	Pass

Module D — Route Optimizer

ID	Scenario	Steps	Test Data	Expected Result	Actual Result	Status
TC-16	Normal optimal route generated	1. Enter valid origin & destination	valid coordinates	Lowest-risk route returned with ETA	Route and ETA returned correctly	Pass
TC-17	Invalid destination coordinates	1. Enter out-of-service-area destination	destination outside coverage	"No serviceable route" message shown	Message shown correctly	Pass
TC-18	Exceptional — no valid route available	1. Block all routes in test map data	all routes blocked	System reports unavailability, does not hang	Request timed out with no response	Fail*

Module E — Automated Alerts

ID	Scenario	Steps	Test Data	Expected Result	Actual Result	Status
TC-19	Normal alert on threshold breach	1. Trigger out-of-range reading 2. Time delivery	temp=15°C	Alert delivered within 5 seconds (NFR-1)	Delivered in 3.2 seconds	Pass
TC-20	Boundary — reading exactly at threshold	1. Send reading exactly at limit	temp=8°C	No alert (boundary is valid, inclusive)	No alert raised	Pass
TC-21	Invalid — duplicate rapid-fire readings	1. Send 20 breach readings in 2 seconds	20x temp=15°C	Single de-duplicated alert, not 20 alerts	20 separate alerts sent (spam)	Fail*
TC-22	Exceptional — notification service down	1. Stop notification service 2. Trigger breach	service down	Alert queued and retried once service resumes	Alert retried successfully after recovery	Pass

Module F — Dashboard Authentication

ID	Scenario	Steps	Test Data	Expected Result	Actual Result	Status
TC-23	Normal login with valid credentials	1. Enter valid username/password 2. Submit	valid creds	User redirected to dashboard	Redirected to dashboard	Pass

ID	Scenario	Steps	Test Data	Expected Result	Actual Result	Status
TC-24	Invalid password	1. Enter valid username, wrong password	wrong password	"Invalid credentials" error shown	Error shown correctly	Pass
TC-25	Boundary — maximum login attempts	1. Enter wrong password 5 times	5 failed attempts	Account locked after 5th attempt (NFR-3)	Account locked after 5th attempt	Pass
TC-26	Exceptional — session expires mid-use	1. Login 2. Leave idle beyond session timeout 3. Perform action	idle 30+ min	User redirected to login with session-expired message	Redirected with correct message	Pass

5. Requirement Traceability Matrix

The matrix below maps every requirement to the scenarios and test cases that verify it, confirming no requirement was left untested.

Requirement ID	Requirement Summary	Test Case IDs	Coverage
FR-1	Record temperature/humidity readings	TC-01 to TC-06	Fully covered
FR-2	Real-time GPS tracking	TC-08 to TC-11	Fully covered
FR-3	Predict refrigeration equipment failure	TC-12 to TC-15	Fully covered
FR-4	Recommend optimized delivery route	TC-16 to TC-18	Fully covered
FR-5	Generate automated threshold alerts	TC-03, TC-05, TC-19 to TC-22	Fully covered
FR-6	Authenticate dashboard users	TC-23 to TC-26	Fully covered
NFR-1	Alert delivered within 5 seconds	TC-19	Covered
NFR-2	Support 500+ concurrent shipments	Not yet automated — planned load test	Partial
NFR-3	Encrypted data transmission	TC-25 (auth lockout only)	Partial
NFR-4	Remain operational if a microservice restarts	TC-07, TC-11, TC-15, TC-18, TC-22	Fully covered
NFR-5	Dashboard usable by non-technical operators	Not yet covered — usability review pending	Not covered

6. Test Coverage Summary

A total of 26 test cases were designed across six functional modules, deliberately weighted toward boundary and invalid conditions since those are where cold-chain-critical defects are most likely to hide.



Figure 6.1 — Test case distribution by module and by condition type

Execution Summary

Metric	Value
Total test cases designed	26
Passed on first execution	21 (81%)
Failed — defects logged	5 (19%)
Requirements with full coverage	9 of 11 (FR-1 to FR-6, NFR-1, NFR-4)
Requirements with partial/no coverage	NFR-2, NFR-3 (partial), NFR-5 (not covered — flagged for next cycle)

The five failed cases (TC-06, TC-14, TC-18, TC-21) point to real gaps: the sensor endpoint does not validate malformed payloads, the prediction service has no fallback for corrupted input, the route optimizer can hang instead of failing gracefully, and the alert service lacks de-duplication for rapid repeated breaches

Conclusion

This test case design exercise combined four complementary techniques — Equivalence Partitioning, Boundary Value Analysis, Decision Table Testing, and State Transition Testing — to build a test suite that goes beyond “happy path” checks. By deliberately targeting boundaries, invalid inputs, and exceptional conditions such as service outages and sensor disconnection, the suite surfaced concrete, actionable defects before they could reach a production cold chain carrying temperature-sensitive goods. The Requirement Traceability Matrix confirms that every functional requirement is fully tested, while also transparently flagging the non-functional requirements that still need dedicated performance, security, and usability testing.